

Laurenz Walter Torni

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About me

Currently I'm a MSc student in mechanical engineering at the Technical University of Denmark, with interdisciplinary interests in data science, structural mechanics and fluid mechanics.

This semester I am taking mainly data science-related courses, including an ongoing project course where I model a elastomer-metal impact force from experimental data with machine learning and explore symbolic regression in Python to derive an analytical expression. I have experience with Python and Fortran and in building experimental test rigs. I am curious about any technical problem, which also led me to teach myself app development.

For my master's thesis, I am looking for a self-funded, full-time six-month project starting around September 2027, ideally in modelling and simulation that combines machine learning with solid mechanics or dynamics.

Experience

Dialogue One A/S — **May 2026 – Present**
Part-Time student job — Copenhagen

- Project Team Lead
  Jul. 2026 – Present
  - Coordinating project execution together with the project manager on a B2B biogas sales project in the sustainable energy sector
  - Continuously conducting market analysis for client presentations and business development
- Student Consultant
  May – Jun. 2026
  - Supporting outreach-focused projects in the renewable energy sector, primarily in B2B solar and energy efficiency
  - Combining research, client communication, and logistical planning to connect business potential with concrete opportunities

Competence Center Tribology — **Apr. 2025 – Jul. 2025**
Bachelor's thesis — Mannheim

- Designed a modular high-vacuum tribological test system using structured engineering methods (VDI 2222/2225/2803)
- Performed mechanical dimensioning and collision simulations in Python and selected vacuum-compatible materials and components
- Created CAD models in Inventor, technical drawings, and cost assessments

Turbo Energy Germany GmbH — **Apr. 2024 – Sep. 2024**
R&D Intern - Simulation / Calculation Engineer — Mannheim

- Created calculation programs and conducted parameter studies in Python and Excel for high-speed turbocharger components, including axial thrust bearings, rotor–shaft press fits, tie rods, and bolted joints in cast housings
- Analyzed thermodynamic cycles for refrigeration systems, heat pumps, and ultra-low-temperature processes in Excel and conducted market research on refrigeration system
- Built a datafile builder interfacing with the company's material database to generate CAD-specific input files for material properties, significantly reducing datafile creation time

Hallstein Maschinenbau GmbH — **Dec. 2021 – Feb. 2022**
Intern (Winter Break) — Hemsbach

- CNC machining (lathe) incl. basic G-code programming
- Welding, surface finishing, and mechanical assembly
- Quality control of precision mechanical components

ZG Raiffeisen Heddesheim **Sep. 2021 – Nov. 2021**
Apprentice Mechanic for a agricultural machinery Heddesheim

- Maintenance and troubleshooting on agricultural machinery and repairs to ensure equipment reliability

Education

Danish Technical University **Aug. 2025 - Present**
Mechanical Engineering Master of Science in Engineering

Eventually relevant projects within my studies:

- FEM code development for the vibrational analysis of a snowboard in Matlab
  Investigating the influence of damping elements on the frequency response and distributed kinetic and potential energies of a Snowboard within the Mindlin theory
- FEM thermoelastic code development in Fortran
  Investigation of the thermo-introduced stresses on a piston / Investigation of the temperature on the eigenfrequency of a slender beam
- Investigation of lateral dynamics and stability of a flexible rotor-bearing system
  Development of a mathematical model with experimental calibration for giving design proposals

The Hong Kong University of Science and Technology **Aug. 2026**
Cultural Exchange for two weeks Summerschool

University of Applied Sciences Mannheim **Mar. 2022 – Jul. 2025**
Mechanical Engineering Bachelor of Science
1.3 GPA, Scale: 1 (very good) - 5 (insufficient)

- *Bachelorthesis:* Development of an Adaptive Pin-on-Disc Test Module for Tribological Investigations in High Vacuum
- *Further research project:* Compared conventional neural networks with Physics-Informed Neural Networks (PINNs) for dynamical system of a double pendulum, reducing prediction error by 90% for a double pendulum and 78% for 1D heat diffusion
- *German and Baden-Württemberg Scholarship*: Due to outstanding academic achievement

ITESM: Tec de Monterrey **Aug. 2023&24 – Dec. 2023&24**
Mechanical Engineering Exchange Semesters

Rudolph Steiner School Mannheim **Sep. 2008 – Aug. 2021**
1.7 GPA, Scale: 1 (very good) - 5 (insufficient) High School

- *Award:* Physics high school award for outstanding performance in the subject of physics

Projects

MyVi Visionboard App
React Native App – Development

- Teaching myself app development with React Native out of general curiosity about coding; the beta-version (IOS) can be tested via Apple Testflight

Gascooling and heatradiation of bioenergy powerplants for Environtec GmbH

- Development of calculation tools in Excel for designing bioenergy powerplants

Skills	**Python & Matlab** Various simulations and data visualizations	**Fortran:** Development of a 2D FEM code	**React Native** Native app development in Java Apple Testflight	**CAD:** Inventor & Creo
Languages	English (C1), IELTS	German (C2)	French (B1)	Spanish (A2)